PI

CREDIT RISK & SCORECARD ENGINEERING · CASE STUDY

LoanLens: Out-of-Time Credit Default Scorecard

BI Scientist • AI/ML Engineer • Python Developer

Leak-free credit default model evaluated on 55,518 closed Prosper loans with strict chronological splitting and an approval policy simulator.

CREDIT RISK VULNERABILITY

The Hidden Leakage Trap in Credit Scoring Models

Many public credit models use columns that only exist after a loan is issued, so they look better than they would on new applications.

Post-Origination Leakage

- Models inadvertently include post-origination columns (`LP_CustomerPayments`, `LP_GrossPrincipalLoss`).
- Artificially inflates AUC to 0.95+ in testing.
- Completely unavailable when evaluating a new applicant.

Random Train-Test Splits

- Splitting loan portfolios randomly mixes past macroeconomic regimes with future loans.
- Model memorizes market conditions rather than credit fundamentals.
- Must enforce a strict chronological Out-of-Time (OOT) holdout.

Lack of Policy Simulation

- Credit risk teams care about approval rates and portfolio yield, not raw AUC.
- Without a cutoff policy simulator, credit committees cannot set risk tolerances.
- Need calibrated probabilities linked to financial yield.

METHODOLOGY

Chronological Out-of-Time Split & Calibrated Scorecard

Trained on 2005–2011 originations (41,638 loans) and tested on future 2011–2014 originations (13,880 loans).

Strict Leakage Exclusion

- Excluded all 14 post-origination payment columns.
- Filter strictly on closed loans (Completed vs ChargedOff).
- Enforced by automated unit tests in CI.

Calibrated GBDT

- HistGradientBoosting with monotonic constraints.
- Isotonic probability calibration.
- 300–850 standardized FICO-style credit score.

Policy Simulator

- Interactive Max-PD cutoff slider.
- Expected portfolio yield accounting for 65% LGD.
- Fair lending approval gap audited across slices.

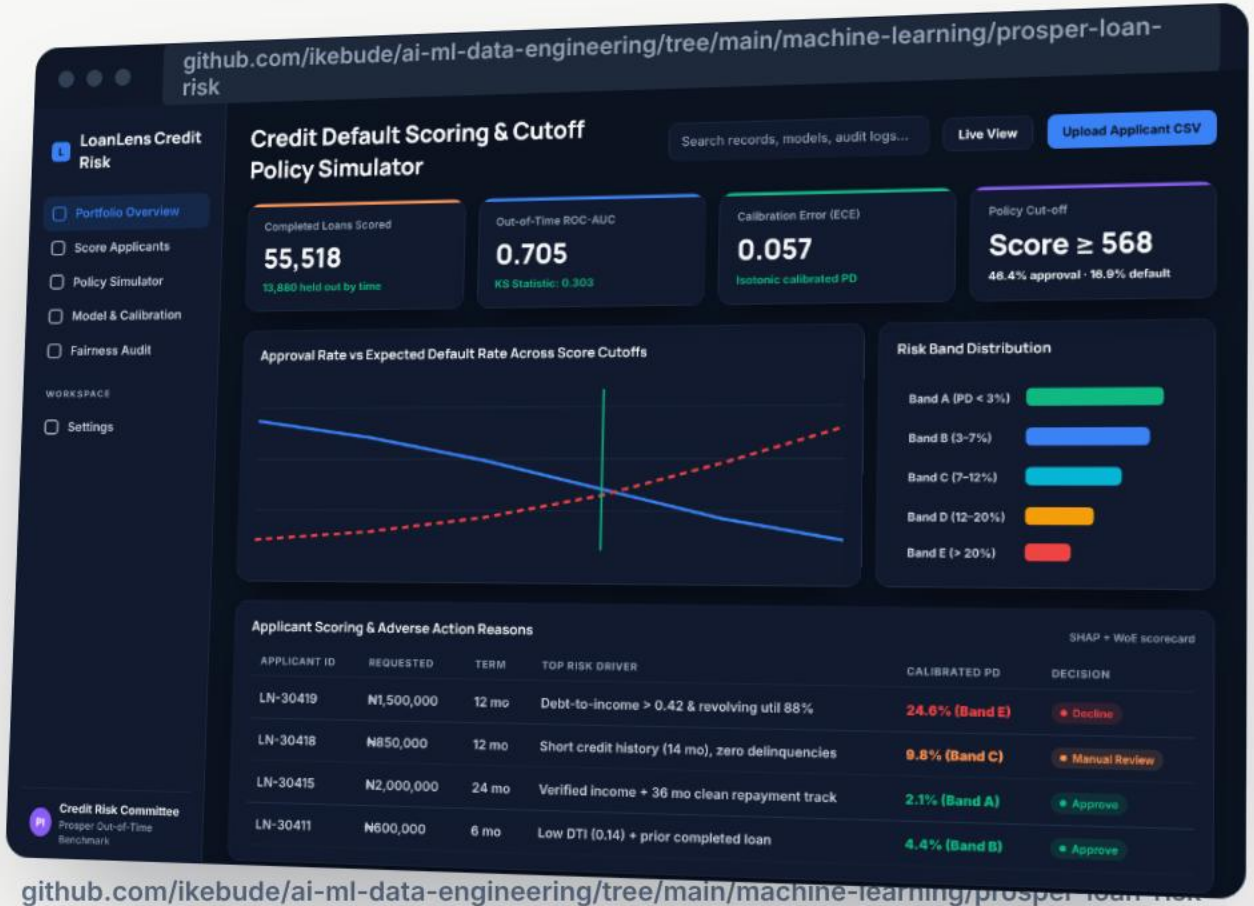
UNDERWRITER INTERFACE

LoanLens Credit Decision Console

Streamlit application simulating approval policies and credit score distributions.

- **Cutoff Policy Simulator:** Adjust cutoff thresholds to visualize trade-offs between approval volume and default rate.
- **Standardized Scorecard (300–850):** Maps calibrated default probability directly into banking credit scores.
- **Fairness Check:** Homeowner vs non-homeowner approval rates differ by 1.8 percentage points.

Credit Risk | Scikit-learn | Streamlit | Prosper Data



github.com/ikebude/ai-ml-data-engineering/tree/main/machine-learning/prosper-loan-risk

OUT-OF-TIME RESULTS

Out-of-Time Generalization on Future Loans

From reports/metrics.json: 13,880 later originations the model never saw.

OOT ROC-AUC	KS Statistic	Approved-Loan Default	Net Portfolio Yield
0.705	0.303	16.9%	+6.61%
95% CI: 0.695–0.714	Class separation	vs 30.7% across all test loans	At 25% max-PD cutoff

Commercial Portfolio Impact

- A 25% Max-PD cutoff (score ≥ 568) approves 46.4% of all applicants.
- Default rate on approved loans is 16.9%, against 30.7% for all test loans.
- Net expected yield +6.61% after a 65% loss-given-default assumption.

Methodological Rigor

- Expected Calibration Error (ECE) of 0.057 confirms probability accuracy.
- Post-origination columns are excluded, and a test enforces it.
- Public Prosper data; a back-test, not a lending decision.

OPEN-SOURCE & REPRODUCIBLE

Honest credit modeling without temporal leakage.

LoanLens bridges the gap between machine learning research and banking credit risk policy.

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Code: github.com/ikebude/ai-ml-data-engineering